M4: Short Answers on Rapidity

Complete Solutions with Teaching, Analysis, and Derivations

PHYS 7502 — Prof. Frantz

Reference Definitions (from Class Slides)

Key Concept

Class (Frantz) definition of rapidity y: defined with respect to the *beam axis* (the z-direction), using the longitudinal component $p_z \equiv p_\parallel$:

$$\boxed{y = \frac{1}{2}\ln\left(\frac{E + p_z}{E - p_z}\right) = \frac{1}{2}\ln\left(\frac{1+\beta_z}{1-\beta_z}\right) = \operatorname{arctanh}(\beta_z)} \tag{1}$$

where $\beta_z = p_z/E = \beta\cos\theta$, with θ the polar angle from the beam.

Pseudorapidity η: a purely geometric/angular quantity (no mass or energy needed):

$$\boxed{\eta = -\ln\left[\tan\left(\frac{\theta}{2}\right)\right] = \frac{1}{2}\ln\left(\frac{|\vec{p}| + p_z}{|\vec{p}| - p_z}\right) = \operatorname{arctanh}\left(\frac{p_z}{|\vec{p}|}\right)} \tag{2}$$

Textbook definition (e.g. Mann Eq. 2.6 p. 41; L&B Eq. 9.43): rapidity of the *Lorentz boost itself*, relating the Lorentz factor to the boost parameter via

$$\gamma = \cosh y, \qquad \beta\gamma = \sinh y, \qquad \beta = \tanh y \tag{3}$$

Part (a): Difference Between Our Definition and the Textbook Definition

Teaching & Derivation

The textbook definition (3) treats rapidity as a property of a *one-dimensional Lorentz boost*. It parametrizes the boost velocity β via $\beta = \tanh y$, so $\gamma = \cosh y$. This implicitly assumes the particle's momentum is *entirely along the boost direction* (i.e. $\theta = 0$ or $180°$, meaning $\vec{p}$ is parallel to $\hat{z}$).
In contrast, the class definition (1) is defined for a particle moving at *any angle* θ relative to the beam. It uses only the *longitudinal projection*: $\beta_z = \beta\cos\theta = p_z/E$. Therefore our definition gives:

$$\tanh y_{\text{class}} = \beta\cos\theta = \beta_z$$

while the textbook gives $\tanh y_{\text{text}} = \beta$. These are equal **only when** $\theta = 0$ **or** $180°$ (particle along the beam line).

Equivalently: The class definition is specifically tailored for collider physics, where one always decomposes momentum into components parallel and transverse to the beam. The textbook definition describes the rapidity of a reference frame (or a particle moving in one dimension). Our definition describes the *longitudinal* rapidity of a particle in 3D, which is naturally tied to the beam axis. This is also the definition used by Thomson's *Modern Particle Physics* §10.9.1.

Connection: For a particle with transverse mass $m_T = \sqrt{m^2 + p_T^2}$, our definition gives:

$$E = m_T\cosh y, \qquad p_z = m_T\sinh y \tag{4}$$

Compare with the textbook: $E = m\cosh y_{\text{text}}$, $|\vec{p}| = m\sinh y_{\text{text}}$. These coincide only if $p_T = 0$ (so $m_T = m$ and $p_z = |\vec{p}|$).

Answer (2 sentences max per sub-question)

The class definition $y = \frac{1}{2}\ln\left(\frac{E+p_z}{E-p_z}\right) = \operatorname{arctanh}(\beta\cos\theta)$ uses only the *longitudinal* (beam-axis) component of the velocity, so it applies to particles at any angle θ. The textbook definition $\gamma = \cosh y$, $\beta = \tanh y$ parametrizes the *full* boost speed and agrees with ours only for particles moving entirely along the beam ($\theta = 0°$ or $180°$, i.e. $p_T = 0$).

Part (b): Numerical Comparison of η vs. y for the Same Particle

Teaching & Derivation

Compare the two definitions side by side:

$$y = \frac{1}{2}\ln\left(\frac{E + p_z}{E - p_z}\right) = \operatorname{arctanh}\left(\frac{p_z}{E}\right) \tag{5}$$

$$\eta = \frac{1}{2}\ln\left(\frac{|\vec{p}| + p_z}{|\vec{p}| - p_z}\right) = \operatorname{arctanh}\left(\frac{p_z}{|\vec{p}|}\right) \tag{6}$$

The **only** difference is that y has E in the denominator while η has $|\vec{p}|$. Since the energy-momentum relation gives $E = \sqrt{|\vec{p}|^2 + m^2}$, we always have $E \geq |\vec{p}|$ with equality when $m = 0$.

Key consequences:

- For **massless particles** (γ, ν, and effectively e at high energy): $E = |\vec{p}|$ exactly, so $y = \eta$ **exactly**.
- For **massive particles at high energy** ($E \gg m$, i.e. $|\vec{p}| \gg m$): $E \approx |\vec{p}|$, so $y \approx \eta$ to excellent approximation, *except* at very forward/backward angles where $p_z \to \pm|\vec{p}|$.
- Quantitatively: at mid-rapidity ($\theta \approx 90°$) even for particles like pions or protons at typical collider energies, $|y - \eta|$ is very small. The difference only becomes significant at very large $|y|$ (very forward/backward), as shown on the class slides.

Practical regime: In most experimental contexts at high-energy colliders, the particles of interest have $p \gg m$, so y and η are **effectively the same** or **almost exactly the same**.

When is the maximum (beam rapidity) value relevant? The maximum pseudorapidity is technically $\eta \to \infty$ as $\theta \to 0$ (beam direction), since $\eta = -\ln[\tan(\theta/2)]$. However, the rapidity y for the beam particle itself defines the practical *maximum interesting* value y_{beam}. For a massless particle (or when $m \to 0$), $\eta_{\text{beam}} \to \infty$. In practice, the beam rapidity y_b sets the boundary: particles are produced with $|y| \lesssim y_b$, and the maximum *pseudorapidity* of produced particles is bounded by the beam pseudorapidity. Therefore y is "preferred" in the sense that it has a well-defined, finite maximum even for the beam particles ($y_b = \operatorname{arctanh}(\beta_{\text{beam}})$), while $\eta_b \to \infty$ for the beam itself.

Answer (2 sentences max per sub-question)

For practical purposes η and y are best described as **effectively the same** (or "almost exactly the same"), because at collider energies $E \approx |\vec{p}|$ for most particles; they differ only at very forward/backward angles for massive particles. Rapidity y is preferred when considering a maximum value, because the beam rapidity $y_b = \operatorname{arctanh}(\beta_{\text{beam}})$ is finite and well-defined, whereas the beam *pseudo*rapidity $\eta_b \to \infty$ since $\theta_{\text{beam}} = 0$.

Part (c): Two Advantages of y or η over θ

Teaching & Derivation

Advantage 1: Lorentz-invariant differences (additivity under boosts). Under a Lorentz boost along the beam axis with velocity β_F, the rapidity transforms as:

$$y' = y + y_{\beta_F}$$

where $y_{\beta_F} = \operatorname{arctanh}(\beta_F)$ is a constant (depending only on the boost, not on the particle). Therefore:

$$\Delta y = y_2 - y_1 = y_2' - y_1' = \Delta y'$$

Rapidity differences are Lorentz invariant. This means particle distributions dN/dy have the **same shape** in every longitudinally boosted frame—they are merely shifted by a constant y_{β_F}.
This is an enormous advantage for comparing fixed-target and collider experiments, or for defining Lorentz-invariant angular sizes of jets ($\Delta R = \sqrt{\Delta y^2 + \Delta\phi^2}$).
The polar angle θ has no such simple transformation under boosts—it mixes with energy in a complicated, particle-dependent way (relativistic aberration).

Advantage 2: Flat particle distributions. Produced-particle distributions dN/dy (and approximately $dN/d\eta$) tend to be roughly **flat** (plateau-like) over a broad range of rapidity in the central region. By contrast, $dN/d\theta$ is extremely peaked near $\theta \approx 0$ and $\theta \approx 180^\circ$ at high energies. The rapidity/pseudorapidity variable "unfolds" the extreme forward/backward peaking caused by the Lorentz boost, spreading the particles out uniformly. This makes it far more practical to bin and display data.

Do these apply to both y and η, or just one?

- **Advantage 1** (Lorentz invariance of Δy) applies **strictly only to y**, not to η. Pseudorapidity η is a purely geometric quantity and does *not* transform additively under boosts (since $|\vec{p}|$ does not transform like E). However, since $y \approx \eta$ for most practical cases, $\Delta\eta \approx \Delta y$ is approximately invariant.
- **Advantage 2** (flat distributions) applies **to both** y and η in practice, since $dN/d\eta \approx dN/dy$ when $m/p_T \ll 1$.

Answer (2 sentences max per sub-question)

(1) Rapidity differences $\Delta y = y_2 - y_1$ are *Lorentz invariant* under longitudinal boosts (since $y' = y + y_F$), whereas θ transforms in a complicated, particle-dependent way; this applies strictly to y only, but approximately to η since $\eta \approx y$ at high energy. **(2)** Particle distributions dN/dy and $dN/d\eta$ are approximately *flat* over a broad central region, whereas $dN/d\theta$ is extremely forward/backward peaked—this advantage applies to both y and η.

Part (d): Why y and η Can Be Interpreted as Radians

Teaching & Derivation

The reasoning comes from evaluating the relationship between η and θ at 90°. From the definition:

$$\eta = -\ln\left[\tan\left(\frac{\theta}{2}\right)\right]$$

we compute the Jacobian:

$$\frac{d\eta}{d\theta} = -\frac{1}{2}\cdot\frac{1}{\tan(\theta/2)}\cdot\frac{1}{\cos^2(\theta/2)} = -\frac{1}{2\sin(\theta/2)\cos(\theta/2)} = -\frac{1}{\sin\theta}$$

At $\theta = 90°$ (i.e. $\eta = 0$, $y = 0$):

$$\left|\frac{d\eta}{d\theta}\right|_{\theta=90°} = \frac{1}{\sin 90°} = 1$$

Therefore, at $\theta = 90°$:

$$\boxed{|d\eta| = |d\theta| \quad (\text{at } \theta = 90°)}$$

Since θ is measured in radians, η has the same numerical scale as radians *at this point.* Now here is the crucial insight: because rapidity differences Δy are **Lorentz invariant**, and because $\Delta y \approx \Delta\eta$, the angular interpretation of a unit of rapidity as "one radian of angle in the rest frame (COM frame)" is preserved by boosts.

That is: a separation $\Delta y = 1$ between two particles corresponds to an angular separation of approximately 1 radian ($\approx 57.3°$) *in the center-of-mass frame*, regardless of what frame you are observing from. This is exactly why jet cone sizes and ΔR values of order unity correspond to "about one radian" of angular opening.

To what extent is this actually true? It is *exactly* true only at $\eta = y = 0$ (i.e. $\theta = 90°$). Away from 90°, the Jacobian $|d\eta/d\theta| = 1/\sin\theta \neq 1$, so one unit of η corresponds to a *different* amount of θ in the lab frame. However, by boost invariance of Δy, the interpretation "$\Delta y = 1$ corresponds to 1 radian in the unboosted frame" carries to all values of y. This is the power of rapidity: it provides a *boost-invariant angular measure.*

Connection to Δy between two particles or two frames: If two particles have $\Delta y = 1$, they are separated by approximately 1 radian in the COM. If two reference frames differ by $\Delta y = y_F$, this corresponds to a relative angular displacement equivalent to y_F radians of COM-frame angular space.

Answer (2 sentences max per sub-question)

At $\theta = 90°$ one can show $|d\eta/d\theta| = 1/\sin\theta = 1$, so $|d\eta| = |d\theta|$ (in radians) exactly at midrapidity; because Δy is Lorentz invariant and $\Delta y \approx \Delta\eta$, this "one unit = one radian in the COM frame" interpretation is carried by boost invariance to all values of y, not just $\theta = 90°$. A rapidity difference Δy between two particles (or equivalently between two directions or two reference frames) thus corresponds to Δy radians of angular separation in the center-of-mass frame, though in the lab frame the corresponding $\Delta\theta$ may be vastly different (e.g. milliradians at forward angles).

Part (e): Why (Pseudo)rapidity Is Especially Advantageous for Hadronic Collisions

Teaching & Derivation

In hadronic collisions (X+h or h+h, such as pp, $p\bar{p}$, Au+Au, ep, etc.), the fundamental hard scattering occurs between **partons** (quarks and gluons), not between the hadrons as a whole. Each parton carries a random fraction x of its parent hadron's 4-momentum, described by the parton distribution functions (PDFs) $f(x, Q^2)$.

The critical consequence: In a hadron-hadron collision (e.g. pp at the LHC), the two colliding partons carry momentum fractions x_1 and x_2, which are generally *different* and fluctuate event by event. This means the parton-parton center-of-mass frame is **boosted** relative to the lab frame by a random, event-by-event longitudinal boost:

$$y_{\text{boost}} = \frac{1}{2} \ln\left(\frac{x_1}{x_2}\right)$$

The partonic CM energy is $\hat{s} = x_1 x_2 s$, and the boost depends on x_1/x_2. Since x_1 and x_2 are drawn from broad distributions, the boost y_{boost} varies randomly from event to event.

Why rapidity helps:

1. **Boost invariance of Δy:** Since the partonic CM frame is randomly boosted, any observable defined in terms of θ would depend on the (unknown, fluctuating) boost. But because Δy is Lorentz invariant, angular separations and distributions expressed in rapidity are **the same regardless of the unknown partonic boost**. You don't need to know x_1 and x_2 to interpret rapidity distributions.

2. **Shape invariance of dN/dy:** The rapidity distribution of produced particles has the same shape in every boosted frame—it just shifts by the (random) constant y_{boost}. When averaged over many events (i.e. over the PDF), this produces a smooth, broad, roughly flat distribution in the central region. In contrast, a distribution in θ would be smeared in a complicated, non-trivial way by the fluctuating boosts.

3. **Invariant cross section:** The combination $E\, d^3\sigma/d^3p = d^3\sigma/(dy\, d^2p_T)$ is Lorentz invariant, making y and p_T the natural variables for presenting cross sections in hadronic collisions.

Contrast with e^+e^-: In e^+e^- collisions, the entire beam momentum participates (no PDFs, no partonic substructure), so the CM frame is *fixed and known* ($y_{\text{boost}} = 0$ in a symmetric collider). Using θ and $\cos\theta$ is perfectly fine there. It is precisely the *hadronic* nature of at least one beam—and the resulting random longitudinal boosts from parton momentum fractions—that makes rapidity indispensable.

Answer (2 sentences max per sub-question)

In hadron collisions, each parton carries a random fraction x of the hadron's momentum, so the parton–parton CM frame is randomly boosted by $y_{\text{boost}} = \frac{1}{2}\ln(x_1/x_2)$ event by event; since Δy is Lorentz-invariant, rapidity distributions and angular separations are unaffected by this unknown, fluctuating boost, unlike distributions in θ. This makes (pseudo)rapidity the natural variable: it absorbs the random partonic boosts into a trivial overall shift, so that dN/dy shapes, jet sizes (ΔR), and invariant cross sections $d^3\sigma/(dy\, d^2p_T)$ are all directly comparable across events and experiments without knowing x_1, x_2.

Summary Table of Key Formulas

Quantity	**Class Definition**	**Textbook Definition**
Rapidity y	$\frac{1}{2}\ln\frac{E+p_z}{E-p_z} = \operatorname{arctanh}(\beta\cos\theta)$	$\operatorname{arctanh}(\beta)$
Pseudorapidity η	$-\ln\left[\tan\frac{\theta}{2}\right] = \operatorname{arctanh}\left(\frac{p_z}{\lvert\vec{p}\rvert}\right)$	—
$y = \eta$ when	$m = 0$ (exactly) or $E \gg m$ (approx.)	—
Boost transform	$y' = y + y_F$ (additive!)	Same
Δy invariance	$\Delta y' = \Delta y$ (Lorentz invariant)	Same
Radians interp.	$\lvert d\eta\rvert = \lvert d\theta\rvert$ at $\theta = 90°$; all y by inv.	—
Transverse mass	$E = m_T\cosh y,\ \ p_z = m_T\sinh y$	$E = m\cosh y,\ \ p = m\sinh y$